



CODECHEF

The CodeChef.com Challenge—your monthly dose of coding puzzles from India's biggest coding contest, now in print!

*H*ello and welcome back to the CodeChef Puzzle Challenge sponsored by *LINUX For You* and Directi! Let's see how many of you can solve this month's puzzle, and find out which of you will be the three lucky winners of our Rs 1,000 cash prize!

Last month's puzzle

A prime number is a number divisible by 1 and itself. How many such numbers exist between 1 and 1,000,000?

The solution

There are 78,498 prime numbers between 1 and 1,000,000.

How did we do it?

If you were to check whether every number

for the given range is a prime number or not, it would have taken very long. A faster way to calculate this would be the Sieve of Eratosthenes. Probabilistic algorithms for testing primality, like the Miller-Rabin test would be even faster.

And the winners for April are...

- Sunish Kumar
- Savitri Patel
- Anuvrat Singh

Puzzle for May

What is the maximum number of bishops that can be placed on a 1,000x1,000 chess board such that no bishop can attack any other bishop?

Looking for a little extra cash?

Solve this month's CodeChef Challenge puzzle and mail your answer to codechef@efyindia.com. Three lucky winners get Rs.1000 in cash! 

About CodeChef: CodeChef.com is India's first, non-commercial, online programming competition, featuring monthly contests in more than 35 different programming languages. CodeChef has been created by Directi, as a way to continuously challenge and engage the developer community. The goal is to provide a platform for practice, competition and improvement as well as enable developers to benchmark their skills against their peers. Log on to www.codechef.com to participate in the May Challenge that takes place from May 1st to the 11th, 2010 to win cash prizes up to Rs.55,000.